	7.2. PROFICIENCY TESTING SCHEME DESIGN AND PRODUCTION PLAN		
	Document no: ASMF 7.2.1		
	Compiled by:	**Sifiso Nyambi** **Laboratory Manager**	
	Approved by:	***Nonhlanhla Msibi*** ***Operations Manager***	

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This master document governs the design, statistical strategy, preparation, testing, and distribution of Proficiency Testing (PT) items for coal analysis. It applies to all PT testing items that fall under the scope of the laboratory.

General Information

Field	Entry
PT Scheme	September GA/HGI 09/2026
Unique Batch Number	September GA/HGI 09/2026
Intended Use	Proficiency Testing Sample & Competency Evaluation and monitoring of staff
Matrix Type	Pulverized Coal (< 212 µm) *OR* Prepared Coal Sized Material (4.75 mm)
Properties to be assigned	ISO Ash, ISO Volatile Matter, Calorific Value, Total Sulphur, Inherent Moisture *OR* Hardgrove Grindability Index (HGI)
Production Start Date	Start of PT round
Production End Date	End of PT round
Expected Participants	10 – 30 Laboratories
Responsible Person	PT Scheme Coordinator
Approved By	Operations Manager

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Materials, Equipment & Verification

A list of materials and equipment to be used in the production of the PT testing item are tabled and listed as follows:

General Analysis:

Item	Description/Specs	Supplier	UoM	Verification Date	Verified By	Serial Number	Qualification Service
Raw Material	Bituminous coal sample		-			-	Approved
Crushers	25,10,3 (mm)	MegaSize	-			202203331 SC, AS8809, A124119; respectively	Cleaned and verified
Pulverizer	212 µm	MegaSize	-			22L90713190645	Cleaned and verified
Platform Scale	0.01 Kg	Darvan Brenco	±0.03 kg			120330150	Calibrated
IM Moisture Oven 50L	Maintain a temperature of 107°C	Darvan Brenco	±1.0 °C			DOV53A/210325/40	Calibrated
Mass pieces	1 to 100 (g)	Micron	±0.00014	April 2025	Micron	OB WS1.1R	Calibrated
Ash Furnace	Maintain 815±10°C for at least 1 hour	Micron (ATS)	±4 °C			ATS/210325/7A	Calibrated
ISO Volatile Matter Furnace	900°C±5°C	Micron (ATS)	±3 °C			ATS/210325/59V	Calibrated
Analytical Balance	0.0010g	Micron	0.0003g			ITA2500477, AS2156	Calibrated
Top Balance	0.1g	Micron	0.3g			AEA90242156, AS00A31	Calibrated
Packaging Material	±150 g	Foamy	-	-	-	-	Cleaned
Homogeneity Drum	-	-	-	Checked upon homogeneity test	-	-	Cleaned

Hardgrove Grindability Index:

Item	Description/Specs	Supplier	UoM	Verification Date	Verified By	Serial Number	Qualification Service
Raw Material	Bituminous coal sample	Lungiles Pyrite	-			-	Approved
Crushers	25mm, 10mm & 4.75mm	MegaSize	-			202203331 SC, AS8809, & A124119	Cleaned and verified
Test Sieves	1mm, 3mm & 4.75mm	Micron	-	Sept 2024 Jan 2025		500401, 500623 & 500628	Verified
Platform Scale	0.01 Kg	Darvan Brenco	±0.03 kg			120330150	Calibrated
Drying oven	Maintain a temperature of 40°C	MegaSize	± 1.0°C			LZ 007/07/10	Calibrated
Mass pieces	1 to 100 (g)	Micron	±0.00014	April 2025	Micron	OB WS1.1R	Calibrated
Ash Furnace	Maintain 815±10°C for at least 1 hour	Micron (ATS)	±4 °C			ATS/210325/7A	Calibrated
Analytical Balance	0.0010g	Micron	0.0003g			ITA2500477, AS2156	Calibrated
Top Balance	0.1g	Micron	0.3g			AEA90242156, AS00A31	Calibrated
Packaging Material	±150 g	Foamy	-	-	-	-	Cleaned
Homogeneity Drum	-	-	-	Checked upon homogeneity test	-	-	Cleaned

NB: The supplier of the materials will vary due to the required CRM quality to be produced. However, it is required that a record of list of suppliers of coal associated which a specific CRM be maintained at the laboratory. This is to ensure that the same CRM will be produced that is consistent with the initial quality produced.
NB: All other equipment used for the preparation and testing; the validation or equipment maintenance are specified in the respective coal testing methods ie. Procedure on sample preparation of Hard coal has detailed how to use crushers and efficiency requirements needed to produce coal size material required.

Processing Plan and Workflow

General Analysis

Step	Description	Equipment used	Environmental Conditions	Acceptance criteria
1	Air drying	-	Atmospheric temperature	The sample should be visibly dry
2	Stage Crushing	25,10,3 mm Crushers	Room temperature	Crusher Efficiency ≥90%
3	Pulverizer	Serial no: 22L90713190645	Room temperature	Pulverizer efficiency ≥95%
4	Homogeneity Drum	-	-	¬ 4 hours
5	Homogeneity Validation (analysis)	Ashing Furnace & Inherent moisture Oven	19-26°C	ANOVA, SSbetween<Check Value
6	Packaging	±150g	Room temperature	-
7	Storage	Storage room	≤ 20°C	Temperature <23°C

Hardgrove Grindability Index

Step	Description	Equipment used	Environmental Conditions	Acceptance criteria
1	Air drying	-	Atmospheric temperature	The sample should be visibly dry
2	Stage Crushing	25,10,3 mm Crushers	Room temperature	Crusher Efficiency ≥90%
3	Screening	Serial no: 22L90713190645	Room temperature	Pulverizer efficiency ≥95%
4	Homogeneity Drum	-	-	¬ 4 hours
5	Homogeneity Validation (analysis)	Ashing Furnace Inherent moisture Oven	19-26 °C	ANOVA, SSbetween<Check Value
6	Packaging	±150g	Room temperature	-
7	Storage	Storage room	≤ 20°C	Temperature <23°C

Homogeneity and Stability Assessment

Assessment Type	Method	No of units	Statistics By	Parameter	Acceptance Criteria
Homogeneity	ASIW001	10	PT Scheme Coordinator	Q-Ash and IM	ANOVA, SSbetween<Check Value
Stability	ASIW001	10	PT Scheme Coordinator	Q-Ash and IM	Reproducibility between homogeneity & stability means < homogeneity check value

Measurement Procedures

Parameter	Method Reference	Equipment	Analyst
Q-Ash	ISO 1171	Furnace	Laboratory Analyst
Volatile Matter	ISO 562	Furnace	Laboratory Analyst
Calorific Value	ISO 1928	Bomb Calorimeter	Laboratory Analyst
Inherent Moisture	SANS 5925	Drying Oven	Laboratory Analyst

NB: The following authorised procedures are to be used:
- AS002IW Procedure on the conditioning of samples,
- AS003IW Procedure for the determination of Ash content
- AS006IW Procedure for the determination of Volatile Matter
- AS004IW Procedure on the determination of Inherent Moisture
- AS007IW Procedure on the Calorific Value determination.

Risk Assessment

Risk Description	Impact	Mitigation strategy
Power outage during analysis	Low	Backup generator
Balance drift	Medium	Verification with calibrated mass pieces at the beginning of every shift.
Sample Swap	High	Prepare one sample at a time, only after packaging has been completed then the next sample can be prepared.

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Risk Management

Record Type	File Name/ Location	Responsible Person	Retention Period
Batch Processing	Form	Laboratory Supervisor	5 years
Analytical Results Sheet	LIMS	Laboratory Supervisor	5 years
Homogeneity Test report and Stability Test report	Final PT report or website	PT Scheme Coordinator	10 years
Calibration Certificates	Technical File	Lab Manager	5 Years

Revision Table

Contents Changed	Reason for Change	Revision	By whom	Date
Addition of HGI production planning	Incomplete system documentation	Revision 01	Keitumetse Maleka	06/10/2026

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07/2025	01	07/2028